Chapter 8. Cluster Sampling

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1 Introduction

For weekly food expenditure in Seoul, element sampling needs a list of all families (rarely available) and scatters interviews over the city. A list of blocks is easy to get, and interviewing every family in a few blocks is cheap.

Definition 8.1

A **cluster sample** is an SRS of clusters of elements, observing every element of a selected cluster.

Cluster sampling is effective when

- a good frame of elements is unavailable or costly, while a frame of clusters is easy to obtain;
- the cost of observation increases with the distance between elements.

Since it is SRS of clusters, the estimators parallel those of Ch. 2, with cluster totals playing the role of observations.

Notation.

N	number of clusters in the population
n	number of clusters selected by SRS
m_i	number of elements in cluster i, $i = 1, \ldots, N$
$\bar{m} = \frac{1}{n}\sum_{i=1}^{n} m_i$	average cluster size in the sample
$M = \sum_{i=1}^{N} m_i$	number of elements in the population
$\bar{M} = M/N$	average cluster size in the population
y_{ij}	value of the jth element in cluster i
$y_i = \sum_{j=1}^{m_i} y_{ij}$	total of cluster i

2 Estimation of the Population Mean

The population mean per element is

$$\mu = \frac{\sum_{i=1}^{N}\sum_{j=1}^{m_i} y_{ij}}{\sum_{i=1}^{N} m_i} = \frac{\sum_{i=1}^{N} y_i}{M},$$

a ratio of the cluster totals y_i to the cluster sizes m_i. Its natural estimator is therefore a ratio estimator (Ch. 5) with $x_i = m_i$.

Theorem 8.2: Cluster estimator of μ

$$\bar{y}_{\text{cl}} = \frac{\sum_{i=1}^{n} y_i}{\sum_{i=1}^{n} m_i}, \qquad \widehat{\text{Var}}(\bar{y}_{\text{cl}}) = \frac{N-n}{Nn\bar{M}^2}\, s_c^2, \qquad s_c^2 = \frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y}_{\text{cl}} m_i)^2,$$

where $\bar{M}$ is replaced by $\bar{m}$ if M is unknown. An approximate 95% interval is $\bar{y}_{\text{cl}} \pm 2\sqrt{\widehat{\text{Var}}(\bar{y}_{\text{cl}})}$.

Derivation: from the ratio estimator

With $x_i = m_i$, $R = \mu$ and $\mu_x = \bar{M}$, Ch. 5 gives

$$\text{Var}(\bar{y}_{\text{cl}}) \approx \frac{1}{\bar{M}^2}\cdot\frac{N-n}{N}\cdot\frac{1}{n}\cdot\frac{1}{N-1}\sum_{i=1}^{N}(y_i - \mu m_i)^2 = \frac{1}{\bar{M}^2}\frac{N-n}{Nn}(S_y^2 + \mu^2 S_m^2 - 2\mu\rho S_m S_y),$$

and replacing population quantities by sample ones yields $\widehat{\text{Var}}(\bar{y}_{\text{cl}})$ and the alternative form

$$\widehat{\text{Var}}(\bar{y}_{\text{cl}}) = \frac{N-n}{Nn\bar{M}^2}(s_y^2 + \bar{y}_{\text{cl}}^2 s_m^2 - 2\bar{y}_{\text{cl}}\hat{\rho} s_m s_y), \qquad s_m^2 = \frac{1}{n-1}\sum_{i=1}^{n}(m_i - \bar{m})^2,$$

where s_y^2 is the sample variance of the cluster totals. Cluster totals nearly proportional to cluster sizes ($y_i \approx \mu m_i$) give a small variance.

3 Estimation of the Population Total

3.1 M known

Since $\tau = M\mu$,

$$\hat{\tau} = M\bar{y}_{\text{cl}} = M\frac{\sum_{i=1}^{n} y_i}{\sum_{i=1}^{n} m_i}, \qquad \widehat{\text{Var}}(\hat{\tau}) = M^2\widehat{\text{Var}}(\bar{y}_{\text{cl}}) = N^2\frac{N-n}{Nn}s_c^2.$$

3.2 M unknown

Often the number of elements is unknown. Then use the mean of the cluster totals,

$$\bar{y}_t = \frac{1}{n}\sum_{i=1}^{n} y_i, \qquad \mathrm{E}(\bar{y}_t) = \frac{1}{N}\sum_{i=1}^{N} y_i = \frac{\tau}{N},$$

which is unbiased for the average cluster total, so $N\bar{y}_t$ is unbiased for τ:

$$\hat{\tau} = N\bar{y}_t = \frac{N}{n}\sum_{i=1}^{n} y_i, \qquad \widehat{\mathrm{Var}}(\hat{\tau}) = N^2\frac{N-n}{Nn}s_t^2, \qquad s_t^2 = \frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y}_t)^2.$$

This SRS of cluster totals ignores m_i; it is worse than $M\bar{y}_{\mathrm{cl}}$ when the totals vary mainly through the sizes.

If all $m_i = m$, then $M = Nm$, $\bar{y}_{\mathrm{cl}} = \bar{y}_t/m$, and the two estimators of τ coincide.

4 Comparison with Simple Random Sampling

Assume equal cluster sizes $m_i = m$, so $M = Nm$ and

$$\bar{y}_{\mathrm{cl}} = \frac{\sum_{i=1}^{n} y_i}{nm} = \frac{1}{nm}\sum_{i=1}^{n}\sum_{j=1}^{m} y_{ij} = \frac{\bar{y}_t}{m}.$$

Let $\bar{y}_i = y_i/m$ be the mean of cluster i. Then

$$\widehat{\mathrm{Var}}(\bar{y}_{\mathrm{cl}}) = \frac{1}{m^2}\widehat{\mathrm{Var}}(\bar{y}_t) = \frac{1}{m^2}\frac{N-n}{Nn}\frac{1}{n-1}\sum_{i}(m\bar{y}_i - m\bar{y}_{\mathrm{cl}})^2 = \frac{N-n}{Nn}\frac{1}{n-1}\sum_{i=1}^{n}(\bar{y}_i - \bar{y}_{\mathrm{cl}})^2.$$

Treat the nm observed elements as an $m \times n$ table with column means $\bar{y}_i$ and grand mean $\bar{y}_{\mathrm{cl}}$:

$$\underbrace{\sum_{i=1}^{n}\sum_{j=1}^{m}(y_{ij} - \bar{y}_{\mathrm{cl}})^2}_{SST} = \underbrace{\sum_{i}\sum_{j}(y_{ij} - \bar{y}_i)^2}_{SSW} + \underbrace{m\sum_{i}(\bar{y}_i - \bar{y}_{\mathrm{cl}})^2}_{SSB},$$

with $MSB = SSB/(n-1)$ and $MSW = SSW/[n(m-1)]$. Then

$$\widehat{\mathrm{Var}}(\bar{y}_{\mathrm{cl}}) = \frac{N-n}{Nnm}\,MSB.$$

If the same nm elements had been drawn by SRS from the Nm elements, with sample variance s^2,

$$\widehat{\mathrm{Var}}(\bar{y}) = \frac{Nm - nm}{Nm}\cdot\frac{s^2}{nm} = \frac{N-n}{Nnm}\,s^2.$$

Remark 8.3: Efficiency of cluster sampling

Compare MSB with s^2: cluster sampling is more efficient when clusters are miniatures of the population (heterogeneous within). Usually neighbours are alike, $MSB > s^2$, and it is chosen for lower cost per element.

5 Selecting the Sample Size

Setting $B = 2\sqrt{\frac{N-n}{Nn\bar{M}^2}s_c^2}$ and solving as in Ch. 2:

- for μ with $\bar{y}_{\text{cl}}$: $n = \dfrac{Ns_c^2}{ND + s_c^2}$, $D = \dfrac{B^2\bar{M}^2}{4}$;
- for τ with $M\bar{y}_{\text{cl}}$: $n = \dfrac{Ns_c^2}{ND + s_c^2}$, $D = \dfrac{B^2}{4N^2}$;
- for τ with $N\bar{y}_t$: $n = \dfrac{Ns_t^2}{ND + s_t^2}$, $D = \dfrac{B^2}{4N^2}$,

with s_c^2, s_t^2 taken from a previous survey or a pilot sample.

6 Estimation of a Population Proportion

Let a_i be the number of elements in cluster i having the characteristic. Then p is the ratio $\sum a_i / \sum m_i$ over the population, and

$$\hat{p}_{\text{cl}} = \frac{\sum_{i=1}^n a_i}{\sum_{i=1}^n m_i}, \qquad \widehat{\text{Var}}(\hat{p}_{\text{cl}}) = \frac{N-n}{Nn\bar{M}^2}s_p^2, \qquad s_p^2 = \frac{1}{n-1}\sum_{i=1}^n (a_i - \hat{p}_{\text{cl}} m_i)^2,$$

with $\bar{M} \approx \bar{m}$ if M is unknown. The sample size for a bound B is $n = Ns_p^2/(ND + s_p^2)$ with $D = B^2\bar{M}^2/4$.

7 Cluster Sampling Combined with Stratification

Books read by 4th and 5th graders: strata are grades, clusters are classes. In stratum ℓ, N_ℓ clusters, $\bar{y}_{t\ell}$ the mean sampled cluster total, $\bar{m}_\ell$ the mean sampled size:

$$\text{average cluster total: } \frac{1}{N}(N_1\bar{y}_{t1} + N_2\bar{y}_{t2}), \qquad \text{average cluster size: } \frac{1}{N}(N_1\bar{m}_1 + N_2\bar{m}_2),$$

$$\bar{y}_c = \frac{N_1\bar{y}_{t1} + N_2\bar{y}_{t2}}{N_1\bar{m}_1 + N_2\bar{m}_2} = \frac{\sum_\ell N_\ell \bar{y}_{t\ell}}{\sum_\ell N_\ell \bar{m}_\ell},$$

a combined ratio estimator. Its estimated variance is

$$\widehat{\text{Var}}(\bar{y}_c) = \frac{1}{\left(\sum_\ell N_\ell \bar{m}_\ell\right)^2}\sum_{\ell=1}^L N_\ell(N_\ell - n_\ell)\frac{s_{t\ell}^2}{n_\ell},$$

with $s_{t\ell}^2$ the sample variance of cluster totals in stratum ℓ: the numerator is $\sum_\ell N_\ell^2 \widehat{\text{Var}}(\bar{y}_{t\ell})$, the denominator the squared estimated number of elements.

8 Cluster Sampling with Probabilities Proportional to Size

8.1 The PPS estimator

Sampling with replacement from $y_1, \ldots, y_N$, unit i drawn with probability z_i each draw ($\sum z_i = 1$).

Theorem 8.4: Hansen–Hurwitz estimator

$$\hat{\tau}_{\text{pps}} = \frac{1}{n}\sum_{i=1}^{n}\frac{y_i}{z_i}$$

is unbiased for τ for any choice of $z_i > 0$, with $\widehat{\text{Var}}(\hat{\tau}_{\text{pps}}) = \frac{1}{n(n-1)}\sum_{i=1}^{n}\left(\frac{y_i}{z_i} - \hat{\tau}_{\text{pps}}\right)^2$.

Proof. Each draw gives an iid $t = y_I/z_I$, $P(I = k) = z_k$, with $\text{E}(t) = \tau$; the mean of n draws is unbiased with variance estimated by s_t^2/n. ■

With $z_i = 1/N$ (equal probability), $\hat{\tau}_{\text{pps}} = \frac{N}{n}\sum y_i = N\bar{y}$, the usual estimator. The variance is small when $z_i \propto y_i$, so we want z_i roughly proportional to y_i.

8.2 Application to clusters

Cluster totals tend to be proportional to cluster sizes, so choose $z_i = m_i/M$. With y_i the cluster total and $\bar{y}_i = y_i/m_i$ the cluster mean,

$$\hat{\tau}_{\text{pps}} = \frac{1}{n}\sum_{i=1}^{n}\frac{y_i}{m_i/M} = \frac{M}{n}\sum_{i=1}^{n}\bar{y}_i, \qquad \hat{\mu}_{\text{pps}} = \frac{\hat{\tau}_{\text{pps}}}{M} = \frac{1}{n}\sum_{i=1}^{n}\bar{y}_i :$$

the estimator of μ is the plain average of the sampled cluster means.

Theorem 8.5

$$\widehat{\text{Var}}(\hat{\tau}_{\text{pps}}) = \frac{M^2}{n(n-1)}\sum_{i=1}^{n}(\bar{y}_i - \hat{\mu}_{\text{pps}})^2, \qquad \widehat{\text{Var}}(\hat{\mu}_{\text{pps}}) = \frac{1}{n(n-1)}\sum_{i=1}^{n}(\bar{y}_i - \hat{\mu}_{\text{pps}})^2.$$

Proof. The $\bar{y}_i$ are iid draws (with replacement), so $\widehat{\text{Var}}(\frac{1}{n}\sum\bar{y}_i) = \frac{1}{n}\cdot\frac{1}{n-1}\sum(\bar{y}_i - \hat{\mu}_{\text{pps}})^2$; multiply by M^2 for the total. ■

Remark 8.6: Equal sizes

If $m_i = m$, then $\hat{\mu}_{\text{pps}} = \frac{1}{n}\sum_i\frac{1}{m}\sum_j y_{ij} = \bar{y}_{\text{cl}} = \bar{y}_t/m$, and since $s_t^2 = \frac{m^2}{n-1}\sum(\bar{y}_i - \bar{y}_{\text{cl}})^2$,

$$\widehat{\text{Var}}(\bar{y}_{\text{cl}}) = \frac{1}{m^2}\frac{N-n}{Nn}s_t^2 = \frac{N-n}{N}\cdot\frac{1}{n(n-1)}\sum(\bar{y}_i - \bar{y}_{\text{cl}})^2,$$

which is $\widehat{\text{Var}}(\hat{\mu}_{\text{pps}})$ up to the fpc (PPS draws are with replacement).

9 Selected exercises

Motivation. The exact design effect $1 + (m-1)\rho$, and the exact PPS variance showing that size-proportional selection removes size variation from the error.

Exercise 8.1 (Design effect of cluster sampling)

Let all N clusters have m elements ($M = Nm$), let μ and $S^2 = \frac{1}{Nm-1}\sum_{i=1}^{N}\sum_{j=1}^{m}(y_{ij}-\mu)^2$ be the element mean and variance, and define the intracluster correlation

$$\rho = \frac{\sum_{i=1}^{N}\sum_{j\neq j'}(y_{ij}-\mu)(y_{ij'}-\mu)}{(m-1)(Nm-1)S^2}.$$

For an SRS of n clusters, $f = n/N$, show that

$$\operatorname{Var}(\bar{y}_{\mathrm{cl}}) = \frac{1-f}{nm}\,S^2 \cdot \frac{Nm-1}{m(N-1)}\,[1+(m-1)\rho],$$

and that the ratio of $\operatorname{Var}(\bar{y}_{\mathrm{cl}})$ to the variance of the mean of an SRS of nm elements is approximately $1+(m-1)\rho$ when N is large.

Solution. [Supplement]

Step 1. With equal sizes, $\bar{y}_{\mathrm{cl}} = \frac{1}{n}\sum_{i\le n}\bar{y}_i$ is the SRS mean of the N cluster means $\bar{y}_i = \frac{1}{m}\sum_j y_{ij}$, whose population mean is μ. By Ch. 2,

$$\operatorname{Var}(\bar{y}_{\mathrm{cl}}) = \frac{1-f}{n}S_b^2, \qquad S_b^2 = \frac{1}{N-1}\sum_{i=1}^{N}(\bar{y}_i-\mu)^2.$$

Step 2. Expanding a square of a sum,

$$\begin{aligned}\sum_{i=1}^{N}(\bar{y}_i-\mu)^2 &= \frac{1}{m^2}\sum_i\Big(\sum_j(y_{ij}-\mu)\Big)^2 = \frac{1}{m^2}\Big[\sum_i\sum_j(y_{ij}-\mu)^2 + \sum_i\sum_{j\neq j'}(y_{ij}-\mu)(y_{ij'}-\mu)\Big]\\ &= \frac{1}{m^2}\Big[(Nm-1)S^2 + (m-1)(Nm-1)S^2\rho\Big] = \frac{(Nm-1)S^2}{m^2}[1+(m-1)\rho].\end{aligned}$$

Step 3. Substituting into Step 1 gives the stated formula.

Step 4. (Design effect.) An SRS of nm of the Nm elements has sampling fraction $nm/(Nm) = f$ and variance $\frac{1-f}{nm}S^2$. The ratio is $\frac{Nm-1}{m(N-1)}[1+(m-1)\rho]$, and $\frac{Nm-1}{m(N-1)} \to 1$ as $N\to\infty$.

With $\rho > 0$, nm cluster elements are worth about $nm/[1+(m-1)\rho]$ independent ones; $\rho = 0.05$ doubles the variance for $m = 21$. Efficiency per element needs $\rho \le 0$. ■

Exercise 8.2 (Exact variance of the PPS estimator)

In the setting of the Hansen–Hurwitz theorem (draws with replacement, selection probabilities $z_k > 0$, $\sum z_k = 1$), show that

(a) $\operatorname{Var}(\hat{\tau}_{\mathrm{pps}}) = \frac{1}{n}\sum_{k=1}^{N} z_k\Big(\frac{y_k}{z_k}-\tau\Big)^2$;

(b) if all $y_k > 0$, the variance is zero iff $z_k = y_k/\tau$ for all k;

(c) $\widehat{\operatorname{Var}}(\hat{\tau}_{\mathrm{pps}})$ of the theorem is unbiased for $\operatorname{Var}(\hat{\tau}_{\mathrm{pps}})$;

(d) for clusters with $z_i = m_i/M$, $\operatorname{Var}(\hat{\tau}_{\mathrm{pps}}) = \frac{M}{n}\sum_{i=1}^{N} m_i(\bar{y}_i-\mu)^2$, where $\bar{y}_i$ is the mean of cluster i.

Solution. [Supplement] Let $t_r = y_{I_r}/z_{I_r}$, $r = 1,\dots,n$, where $I_1,\dots,I_n$ are the independent draws, $P(I_r = k) = z_k$. Then $\hat{\tau}_{\mathrm{pps}} = \bar{t}$.

(a) The t_r are iid with mean τ (proof of the theorem) and variance $V_1 = \mathrm{E}(t_1-\tau)^2 = \sum_k z_k(y_k/z_k-\tau)^2$. Hence $\operatorname{Var}(\bar{t}) = V_1/n$.

(b) $V_1 = 0$ iff $y_k/z_k = \tau$ for all k, i.e. $z_k = y_k/\tau$; infeasible (needs the y_k), but variance is small for z_k nearly $\propto y_k$.

(c) For iid t_r the sample variance $s_t^2 = \frac{1}{n-1}\sum_r (t_r - \bar{t})^2$ is unbiased for V_1 (the with-replacement case of Ch. 2). Hence $\mathrm{E}[s_t^2/n] = V_1/n = \mathrm{Var}(\hat{\tau}_{\mathrm{pps}})$, and s_t^2/n is the $\widehat{\mathrm{Var}}(\hat{\tau}_{\mathrm{pps}})$ of the theorem.

(d) With y_i the cluster total, $y_i/z_i = My_i/m_i = M\bar{y}_i$ and $\tau = M\mu$. By (a),

$$\mathrm{Var}(\hat{\tau}_{\mathrm{pps}}) = \frac{1}{n}\sum_i \frac{m_i}{M} M^2(\bar{y}_i - \mu)^2 = \frac{M}{n}\sum_i m_i(\bar{y}_i - \mu)^2.$$

Only cluster *means* vary here; $N\bar{y}_t$ is driven by cluster *totals* $y_i = m_i\bar{y}_i$, which vary also through the sizes. ■